A CASE-STUDY BASED REPORT

ON

Customer Churn prediction Using Random Forest and XGBoost

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

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Signature of the Team members

1. Abstract

Customer churn is one of the major challenges faced by businesses that provide services to customers. It occurs when customers stop using a company's products or services and move to another provider. Losing customers can negatively affect business growth and revenue. Therefore, identifying customers who are likely to leave is important for improving customer retention.

The project, **"Customer Churn Prediction Using Random Forest and XGBoost,"** aims to apply machine learning techniques to predict customer churn and understand the factors influencing customer decisions. The system considers customer-related attributes such as tenure, contract type, monthly charges, payment method, and services used. The dataset is prepared through data preprocessing, followed by Exploratory Data Analysis (EDA) to understand customer patterns and identify potential churn factors.

Two machine learning algorithms, Random Forest and XGBoost, are selected to learn from historical customer data and predict the likelihood of churn. Their performance can be evaluated and compared using appropriate classification metrics, including accuracy, precision, recall, F1-score, and confusion matrices. The project also proposes integrating the trained model into a web-based application that allows users to enter customer details and view churn predictions.

Overall, this project demonstrates the application of machine learning to a real-world business problem. By identifying customers who may be at risk of leaving, the proposed solution can support timely business decisions, improve customer satisfaction, and strengthen customer-retention strategies.

2. Introduction

Customer retention is an important factor in the growth and success of modern businesses. Organizations invest significant time and resources in attracting new customers and maintaining relationships with existing ones. However, customers may discontinue a service because of pricing, service quality, changing requirements, dissatisfaction, or better alternatives offered by competitors.

Customer churn prediction uses historical customer information and machine learning techniques to identify customers who are more likely to discontinue a service. Instead of relying entirely on manual analysis, businesses can use predictive models to identify patterns associated with customer churn and make informed decisions.

This project applies two machine learning algorithms, Random Forest and XGBoost, to the customer churn prediction problem. The work includes dataset preparation, exploratory data analysis, model development, performance evaluation, and a proposed web-based interface for displaying predictions.

2.1 Background

Customer churn occurs when a customer stops using a company's product or service. It is particularly important in industries that depend on subscriptions or long-term customer relationships, including telecommunications, banking, insurance, and digital service providers.

A high churn rate can reduce revenue, increase customer acquisition costs, and affect long-term business growth. Historical customer data can provide information about customer behaviour, service usage, payment patterns, contract preferences, and other characteristics that may be associated with churn.

Machine learning algorithms can identify relationships between these attributes and previously observed churn outcomes. Random Forest combines predictions from multiple decision trees, whereas XGBoost builds an ensemble of trees sequentially to improve predictive performance. These approaches can be applied to classification tasks in which the target variable indicates whether a customer has churned.

2.2 Motivation for the Project

The motivation for this project is the need for a data-driven approach to identifying customers who may discontinue a service.

The major motivations are:

- To understand the factors associated with customer churn.
- To use historical customer data for predictive analysis.
- To apply and compare Random Forest and XGBoost classification algorithms.
- To identify customer groups that may require additional attention.
- To explore how machine learning predictions can support customer-retention decisions.
- To demonstrate how a predictive model can be integrated into a user-friendly application.

The project also provides practical experience in data preprocessing, exploratory data analysis, machine learning, model evaluation, and application development.

2.3 Overview of the Work Carried Out

The project is organized into the following stages:

1. **Dataset collection:** Obtain a customer churn dataset in CSV format containing customer attributes and churn labels.
2. **Data preprocessing:** Examine the dataset, handle missing or inconsistent values where necessary, and prepare categorical and numerical features for modelling.
3. **Exploratory Data Analysis:** Study customer characteristics, churn distribution, and relationships between input attributes and the target variable.
4. **Algorithm selection:** Select Random Forest and XGBoost as the classification algorithms.
5. **Model development:** Prepare training and testing data and train the selected models.
6. **Model evaluation:** Evaluate and compare model performance using suitable classification metrics.
7. **Prediction interface:** Plan or develop a web-based application that accepts customer information and displays a churn prediction.
8. **Interpretation:** Examine the results and discuss how predictions may assist customer-retention strategies.

The actual implementation status of each stage should be reported according to the work completed.

2.4 Scope and Responsible-Use Statement

The scope of this project is limited to predicting customer churn using structured customer data and supervised machine learning algorithms. The project focuses on data preparation, exploratory analysis, model training, evaluation, and the possible integration of predictions into a web-based application.

The proposed system is intended to support decision-making rather than replace human judgement. A churn prediction indicates an estimated likelihood based on patterns learned from historical data; it does not guarantee that a customer will leave.

Customer information should be handled responsibly. Personal or confidential information should be protected, and only necessary data should be used. The model should be evaluated for errors and potential bias before its predictions are used in business decisions. Predictions should not be used to unfairly disadvantage customers or automatically determine access to services.

3. Problem Statement

Businesses that depend on customer relationships often face the challenge of retaining customers who may discontinue their services. Identifying such customers in advance is difficult when customer information is large, complex, and contains many interacting attributes.

Traditional manual analysis may not efficiently identify all patterns associated with customer churn. Consequently, businesses require a systematic approach that uses historical customer information to estimate churn risk and understand the factors associated with customer departure.

The problem addressed by this project is to develop and evaluate a machine learning-based classification approach using Random Forest and XGBoost to predict whether a customer is likely to churn. The study also considers how the resulting predictions could be presented through a user-friendly application to support customer-retention planning.

3.1 Case-Study Scenario

Consider a service provider that has a large number of customers with different contract types, service usage patterns, payment methods, monthly charges, and customer-tenure periods. Some customers continue using the service, while others discontinue it.

The provider wants to identify customers who may be at risk of leaving so that appropriate retention measures can be considered. Analysing every customer manually may be time-consuming and may not reveal complex relationships among customer attributes.

In this case study, historical customer data containing churn outcomes is used to train classification models. Random Forest and XGBoost learn patterns from the available data and estimate whether a customer belongs to the churn or non-churn category. The results can then be evaluated to determine how effectively the models identify customers who churn.

3.2 Central Case-Study Question

The central question of this project is:

How effectively can Random Forest and XGBoost use historical customer information to predict customer churn, and how can the resulting predictions help businesses identify customers who may require retention efforts?

The study also considers the following supporting questions:

- Which customer attributes are associated with churn in the selected dataset?
- How does preprocessing influence the quality of the input data?
- How do Random Forest and XGBoost compare using appropriate classification metrics?
- How can churn predictions be presented in an understandable and useful format?

3.3 Formal Problem Definition

Customer churn prediction is formulated as a supervised binary classification problem.

Let the customer dataset be represented as:

$D = \{(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)\}$

where:

- n represents the number of customer records.
- x_i represents the input features for customer i.
- y_i represents the observed churn label.

The target variable is defined as:

- y = 1: The customer churned.
- y = 0: The customer did not churn.

The input feature vector may contain customer tenure, contract type, monthly charges, payment method, and service-related attributes, depending on the selected dataset.

The objective is to learn a classification function:

$\hat{y} = f(x)$

where f represents the trained machine learning model, x represents the customer attributes, and $\hat{y}$ represents the predicted churn class. If the model supports probability estimation, it may also produce an estimated churn probability.

The study develops Random Forest and XGBoost models and compares their predictive performance on a held-out test dataset. The final model selection should be based on the evaluation results and the requirements of the application.

3.4 Objectives that Guide the Study

The primary objective is to develop a machine learning-based approach for predicting customer churn using Random Forest and XGBoost.

The specific objectives are:

1. To obtain and understand a structured customer churn dataset.
2. To preprocess the data and prepare the features for machine learning.
3. To conduct EDA to examine customer characteristics and churn patterns.
4. To implement Random Forest and XGBoost classification models.
5. To evaluate model performance using accuracy, precision, recall, F1-score, and confusion matrices, as appropriate.
6. To compare the models and identify the more suitable model based on evaluation results.
7. To investigate the importance of customer attributes in the predictions.
8. To explore or develop a web-based interface for displaying churn predictions.
9. To explain the practical usefulness and limitations of the proposed approach.

3.5 Assumptions and Limitations

Assumptions

1. The dataset contains customer-related attributes and a churn target label.
2. The recorded customer information is sufficiently representative of the population to which predictions may be applied.
3. The churn labels are reasonably accurate and consistently defined.
4. Training and testing records are separated appropriately to evaluate performance on unseen data.

5. Relationships learned from historical data may provide useful information about future churn, provided the underlying patterns remain reasonably stable.

Limitations

1. The prediction quality depends on the quality, size, and representativeness of the dataset.
2. Missing values, class imbalance, noisy observations, or incorrect labels may affect model performance.
3. Customer behaviour can change over time, causing the model's performance to decline.
4. The models identify statistical patterns and do not necessarily establish the causes of churn.
5. False-positive predictions may result in unnecessary retention efforts, while false negatives may cause potential churners to be overlooked.
6. Performance measured on a test dataset may not exactly reflect performance in a real business environment.
7. A web-based interface requires additional implementation, testing, and integration beyond training the machine learning models.

4. Dataset and Exploratory Data Analysis

The dataset is the foundation of the customer churn prediction project. It provides the historical information used to understand customer behaviour, identify patterns, and train the classification models.

Exploratory Data Analysis is used to examine the structure, quality, distribution, and relationships within the data before model development. This stage helps identify potential data-quality issues and understand how customer attributes relate to the churn target.

The actual dataset name, record count, feature count, and EDA findings should be added after inspecting the dataset used in the implementation.

4.1 About the Dataset

The project requires a structured customer churn dataset in CSV format. A telecommunications customer churn dataset is one possible source, consistent with the project abstract, but the exact dataset must be confirmed from the file used in the implementation.

The dataset may contain the following attributes:

Attribute	Description

Customer tenure	Duration for which the customer has used the service
Contract type	Type or duration of the customer's contract
Monthly charges	Amount charged to the customer each month
Total charges	Total amount charged over the recorded customer period
Payment method	Method used to pay for the service
Services used	Services subscribed to by the customer
Churn	Target variable indicating whether the customer left

The exact columns and their meanings depend on the selected dataset. The churn column is used as the target variable, while the remaining suitable attributes are used as input features.

Before modelling, the dataset should be examined to determine its dimensions, data types, missing values, duplicate records, categorical variables, numerical variables, and class distribution.

Dataset details to complete after inspection:

- Dataset name: [Enter the actual dataset name]
- Dataset source: [Enter the source]
- Number of records: [Enter the actual number]
- Number of columns: [Enter the actual number]
- Target column: [Enter the exact churn column name]
- Churn classes: [Enter the actual class labels]

4.2 Purpose of EDA

Exploratory Data Analysis is performed to understand the dataset before applying machine learning algorithms.

The main purposes of EDA are:

- **Understanding the dataset structure:** Determine the number of rows and columns, feature names, and data types.
- **Checking data quality:** Identify missing values, duplicate records, and inconsistent values.

- **Understanding numerical features:** Examine the distributions of tenure, monthly charges, and total charges.
- **Examining categorical features:** Understand the distribution of contract types, payment methods, and services used.
- **Studying churn distribution:** Determine the number or proportion of customers in each churn category.
- **Identifying relationships:** Examine how customer attributes vary between churned and non-churned customers.
- **Supporting preprocessing decisions:** Identify the transformations required before model training.
- **Supporting model interpretation:** Identify patterns worth investigating during feature-importance analysis.

Suitable visualizations include count plots, histograms, box plots, bar charts, and correlation heatmaps where appropriate.

4.3 Preparation Steps Actually Performed

The final report must describe the preprocessing steps that were genuinely performed in the project. The following is a preparation framework to use when documenting the implementation.

Dataset loading: Load the CSV file using Pandas.

- **Initial inspection:** Examine the dataset using operations such as head(), shape, info(), and describe().
- **Missing-value inspection:** Check each column for missing values and decide how to handle them where necessary.
- **Duplicate inspection:** Identify duplicate records and remove them only when justified.
- **Data-type verification:** Confirm that numerical and categorical columns have appropriate data types.
- **Target preparation:** Convert the churn label into the format required by the classification models, if necessary.
- **Categorical encoding:** Encode categorical features using a suitable method, such as one-hot encoding, when required by the model pipeline.
- **Feature and target separation:** Separate the input variables from the churn target.
- **Training and testing split:** Divide the data into training and testing sets, preferably using stratification when appropriate for the class distribution.
- **Leakage prevention:** Fit preprocessing transformations using training data only and apply the learned transformations to test data.

The exact steps depend on the dataset and code. For example, if the dataset contains no missing values, missing-value imputation should not be reported as a performed step. Similarly, scaling should only be documented if it was actually required and applied.

4.4 EDA Exploration – Implementation

EDA can be implemented in Python using Pandas, NumPy, Matplotlib, and Seaborn.

The following analyses are suitable for the project:

A. Dataset overview

Inspect the first few records, dimensions, column names, data types, and summary statistics to understand the structure of the dataset.

B. Missing values and duplicates

Calculate the number of missing values in each column and inspect duplicate records. Use the results to determine whether cleaning is necessary.

C. Churn distribution

Create a count plot or bar chart showing the number of customers who churned and those who did not. This helps determine whether the target classes are balanced or imbalanced.

D. Numerical feature analysis

Plot the distributions of numerical attributes such as tenure, monthly charges, and total charges. Histograms can show distributions, while box plots can help identify unusual values and differences between churn categories.

E. Categorical feature analysis

Use count plots or churn-rate comparisons to explore attributes such as contract type, payment method, and services used. These plots can reveal differences in customer behaviour across categories.

F. Feature relationships

Examine relationships between numerical attributes using correlation analysis where appropriate. A correlation heatmap can summarize linear associations between numerical variables, but it does not establish causation.

G. Churn-related patterns

Compare the attributes of churned and non-churned customers. Any observed patterns should be recorded with reference to the actual plots and numerical results.

All figures included in the final report should be generated from the project's real dataset. Each figure should have a number, title, and short explanation of the observation it supports.

4.5 Summary of EDA Findings

The EDA findings should summarize the actual observations obtained from the dataset rather than assumptions about customer behaviour.

The following items can be used to prepare the summary:

Dataset quality: State whether missing values, duplicates, or inconsistent data types were found and how they were handled.

- **Churn distribution:** Report the actual number and percentage of churned and non-churned customers.
- **Customer tenure:** Describe any observed differences in tenure between churn categories.
- **Contract type:** Explain any patterns observed between contract types and churn.
- **Monthly charges:** Summarize the observed relationship between monthly charges and churn.
- **Payment method and services:** Report any meaningful differences identified in the visualizations.
- **Feature relationships:** Mention relevant relationships or potential redundancy among input variables.
- **Modelling implications:** Explain how the observations influenced preprocessing, evaluation metrics, or model development.

The final findings should be written after the EDA code has been executed and the results have been examined. No specific churn rate or feature relationship should be claimed without supporting results.

5. Algorithm Selection and Implementation

This project selects Random Forest and XGBoost for customer churn prediction. Both are ensemble learning techniques that combine multiple decision trees, but they construct their ensembles differently.

The algorithms are suitable candidates for structured customer data because they can learn nonlinear relationships and interactions among customer attributes. Their actual suitability must be established through experimental evaluation on the selected dataset.

5.1 Random Forest

Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions to perform classification.

During training, each tree is built using a bootstrap sample of the training data and considers a subset of features when selecting splits. For classification, the predictions of the individual trees are combined, typically through majority voting or an equivalent probability-based decision.

Working procedure:

1. Prepare the training dataset.
2. Select suitable Random Forest parameters, such as the number of trees and maximum tree depth.
3. Train multiple decision trees using the training data.
4. Generate predictions by combining the tree outputs.
5. Evaluate the model using the test dataset.
6. Examine feature importance if required.

Advantages:

- Can learn nonlinear relationships.
- Handles a mixture of numerical and encoded categorical features.
- Reduces reliance on a single decision tree.
- Can provide useful feature-importance estimates.

Limitations:

- May require substantial memory and computation for large ensembles.
- Its predictions may be less interpretable than those of a single decision tree.

- Feature importance requires careful interpretation and does not establish causation.

5.2 XGBoost

XGBoost, or Extreme Gradient Boosting, is an optimized gradient-boosting algorithm. It builds an ensemble of decision trees sequentially, with each new tree helping improve the model's predictions by optimizing an objective function.

The training process incorporates gradient information and regularization to help control model complexity.

Working procedure:

1. Prepare the training dataset and target labels.
2. Select suitable parameters, including the number of boosting rounds, learning rate, and tree depth.
3. Train the initial boosting model.
4. Add sbsequent trees to improve the objective function.
5. Generate predictions on the test dataset.
6. Evaluate performance and examine feature importance if required.

Advantages:

- Can learn complex relationships in structured data.
- Supports regularization to help control overfitting.
- Provides parameters for controlling model complexity and learning.
- Can offer strong predictive performance on tabular classification tasks.

Limitations:

- Requires careful parameter selection and evaluation.
- May overfit if model complexity is not controlled.
- Can require more tuning than a basic classification model.
- Feature-importance measures do not independently establish causal relationships.

5.3 Model Training and Evaluation

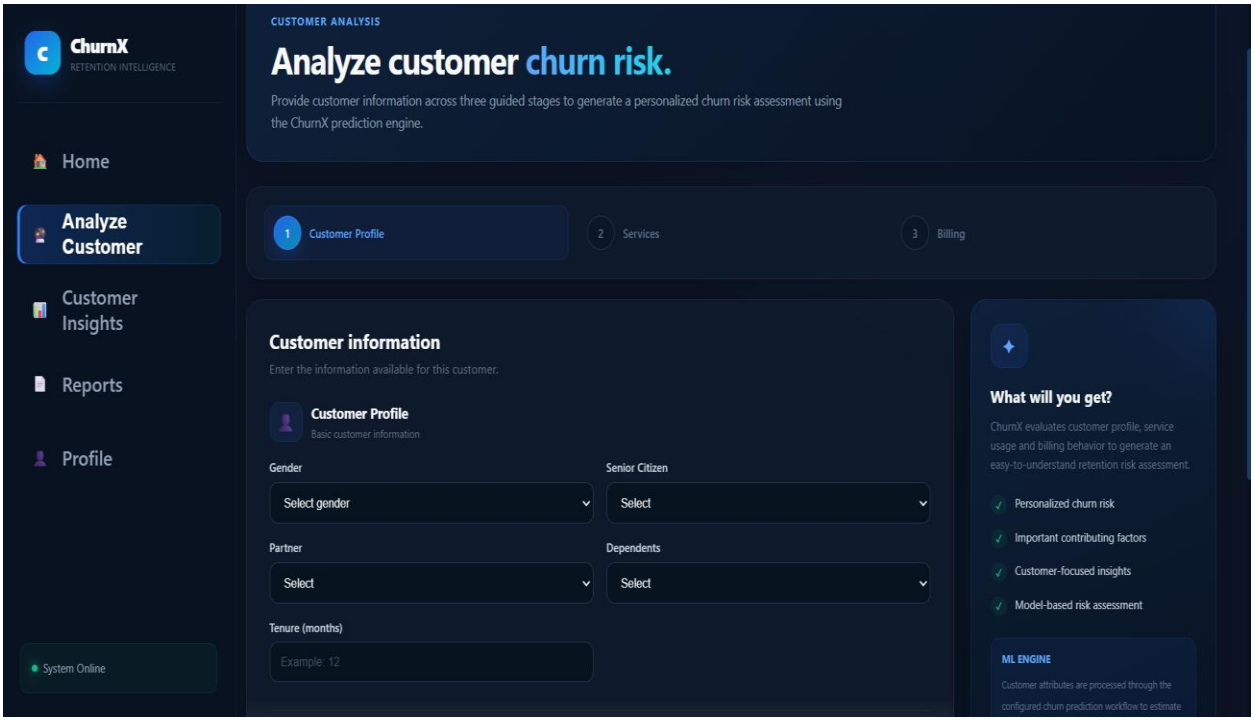
The dataset should be divided into training and testing sets before model development. The training set is used to learn patterns, while the test set is reserved for evaluating predictions on data not used to fit the models.

Both algorithms should be evaluated using the same test split and comparable preprocessing. Hyperparameter tuning, if performed, should use a validation set or cross-validation without using the test set to select the final model.

The following metrics can be used:

- **Accuracy:** The proportion of all predictions that are correct.
- **Precision:** The proportion of customers predicted to churn who actually churned.
- **Recall:** The proportion of actual churners correctly identified.
- **F1-score:** The harmonic mean of precision and recall.
- **Confusion matrix:** Shows true positives, true negatives, false positives, and false negatives.

For customer churn prediction, recall and precision are particularly important because businesses may want to identify as many potential churners as possible without generating too many unnecessary retention interventions.



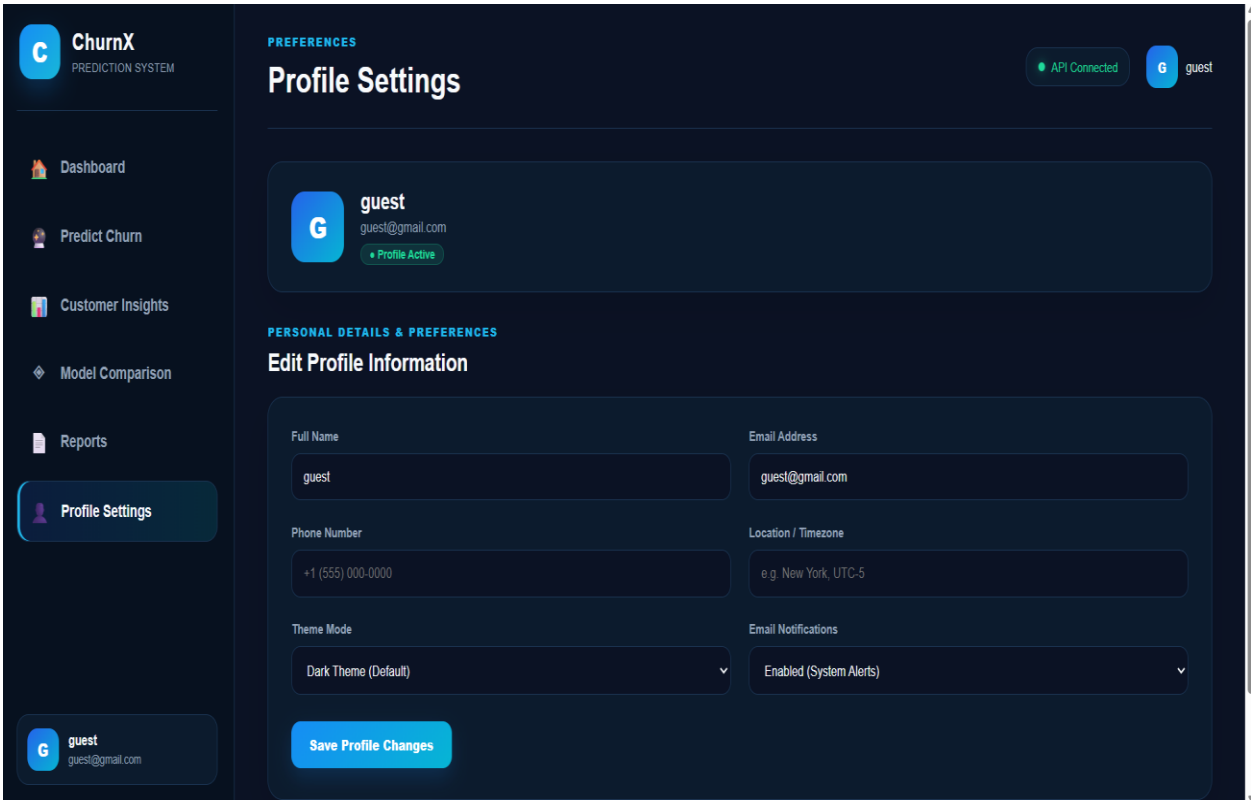
5.4 Model Comparison

Random Forest and XGBoost should be compared using their measured evaluation results. Accuracy alone may not be sufficient, especially if the dataset has an imbalanced churn distribution.

A comparison table can be completed after running the implementation:

Evaluation metric	Random Forest	XGBoost
Accuracy	[Actual result]	[Actual result]
Precision	[Actual result]	[Actual result]
Recall	[Actual result]	[Actual result]
F1-score	[Actual result]	[Actual result]

The final model should be selected according to the measured results, the importance of identifying churners, the acceptable false-positive rate, and the intended application requirements. Neither algorithm should be declared superior before the evaluation has been completed.



5.5 Proposed Web-Based Application

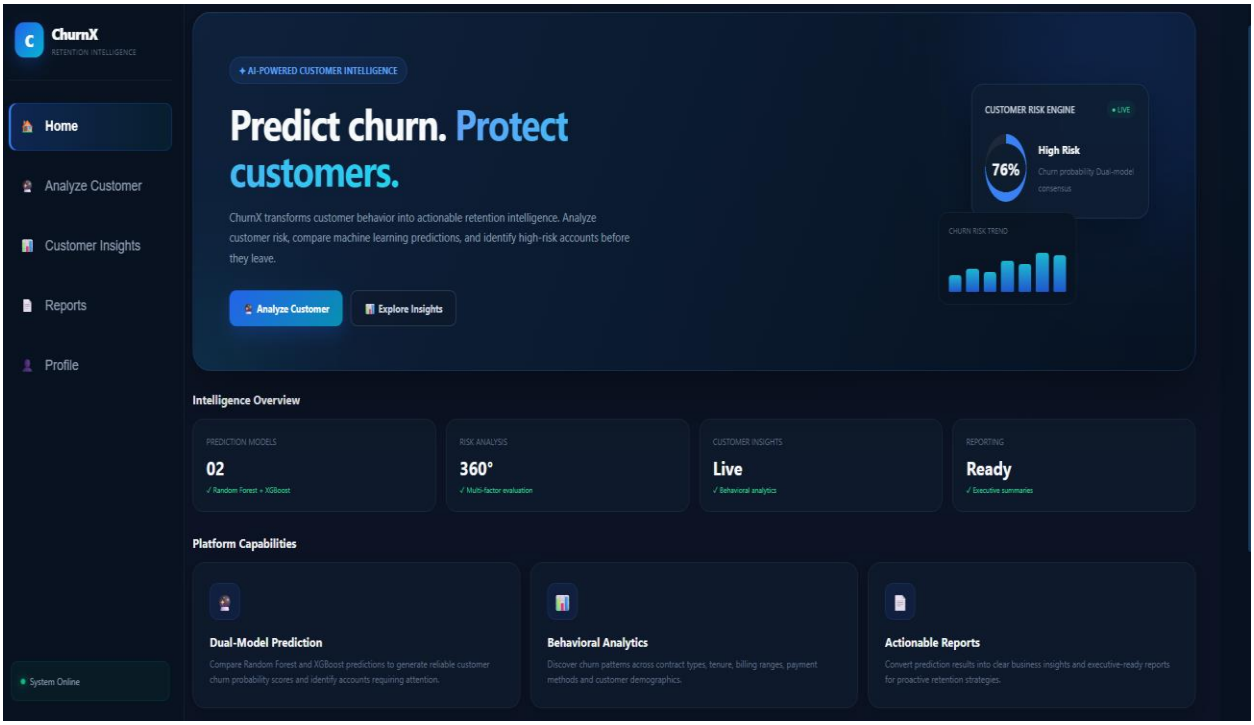
The project abstract proposes integrating the trained model into a web-based application to make churn prediction more accessible.

The application may allow a user to enter relevant customer attributes, submit the information to the trained model, and view the predicted churn category. If the model provides a calibrated or otherwise appropriately interpreted probability, the application may also display an estimated churn risk.

- Potential components include:
- A customer-information input form.
- Input validation and preprocessing.
- Integration with the trained prediction model.
- A prediction result display.
- A summary of relevant model insights, where supported.

The web application should use the same preprocessing transformations as the trained model. It should also communicate that predictions are estimates rather than guarantees.

The application should be described as proposed until the interface and model integration have actually been implemented and tested.



6. Conclusion

Customer churn prediction is an important application of machine learning that can help businesses understand customer behaviour and identify customers who may discontinue their services. This project focuses on developing a classification-based approach using Random Forest and XGBoost.

The planned workflow includes dataset inspection, data preprocessing, Exploratory Data Analysis, model training, evaluation, and comparison. The study also considers how the resulting predictions could be integrated into a web-based application to provide an accessible interface for customer churn analysis.

The effectiveness of the proposed approach depends on the quality of the dataset, appropriate preprocessing, reliable evaluation, and the ability of the trained models to generalize to unseen customers. A fair comparison of Random Forest and XGBoost using suitable metrics can help determine which model is more appropriate for the selected dataset and business objective.

Overall, the project demonstrates how machine learning can be applied to a practical business problem. When developed and evaluated responsibly, a customer churn prediction system can support data-driven retention strategies, help organizations prioritize customer-service efforts, and contribute to improved customer relationships. The final conclusions about model performance and observed churn factors should be based on the actual experimental results obtained during implementation.

Signature of the Team members